Problem 03: AI-Powered Medical Decision-Support Using Clinical Notes, Medical Images, and Patient History

Domain: Healthcare AI | Multimodal Deep Learning | NLP
Difficulty: Hard

Problem

Medical information is spread over radiology images, notes, labs and histories. Reviewing each separately makes it hard to spot patterns and prioritize cases.

Objective

Build a multimodal prototype combining at least two data modalities (e.g., chest X-ray + clinical text/structured data) to support preliminary case review.

Users

Clinicians, radiology triage teams (as a research prototype).

Core Requirements

1. Accept medical images and structured clinical data.
2. Extract information from clinical notes.
3. Image model: pretrained or fine-tuned.
4. Fuse image features with structured data or text.
5. Predict selected abnormalities or risk categories.
6. Show contributing features.
7. Structured case summary with supporting evidence.
8. Flag missing information and uncertain predictions.
9. Evaluate on a held-out set.
10. Clinician-facing dashboard.

Advanced Challenges

- Early vs late fusion.
- Missing modalities at inference time.
- Class imbalance and noisy labels.
- Uncertainty calibration.
- Patient-level split (no patient in both train and test).
- Cross-dataset robustness.
- Multimodal vs unimodal baselines.

Suggested Datasets

IU X-Ray / Open-i (open, with reports), NIH ChestX-ray14, CheXpert, MIMIC-CXR + MIMIC-IV (credentialed PhysioNet access).

Suggested Tech

PyTorch, MONAI, Hugging Face Transformers (ClinicalBERT/BioBERT), scikit-learn, Grad-CAM, FastAPI, Streamlit.

Reference Architecture

DICOM/PNG loader → CNN/ViT encoder; note text → clinical text encoder / NER; tabular → MLP; fusion layer → multi-label head + MC-dropout/ensemble uncertainty → explanation (Grad-CAM, SHAP) → summary generator constrained to extracted facts.

Evaluation

Per-class AUROC, AUPRC, sensitivity at fixed specificity, ECE, performance with missing modalities, ablation vs unimodal baselines.

Expected Deliverable

Dashboard: predictions, heatmaps, relevant clinical facts, feature contributions, uncertainty, structured summary.

Safety Note

Research prototype only. Not a diagnostic system; must not replace professional judgment.

Pitfalls

Same-patient leakage; label leakage from report text containing the diagnosis; ignoring de-identification and data-use terms.

Suggested Timeline (48h)

0-8h data + split; 8-20h image baseline; 20-30h text/tabular + fusion; 30-40h uncertainty + explanations; 40-48h dashboard.

Standard Judging Rubric (applies to every problem)

Criterion	Weight
Problem understanding	10%
Technical depth	20%
Data engineering (quality, leakage prevention, reproducibility)	10%
Model performance (metrics, baselines, honest evaluation)	15%

Criterion	Weight
System architecture	15%
Working prototype	10%
Explainability and responsible AI	10%
Innovation and scalability	5%
Presentation and demo	5%

Mandatory Submission Checklist

- ☐GitHub repository with setup instructions
- ☐Working prototype or recorded demo
- ☐Documented system architecture
- ☐Dataset sources and preprocessing methodology
- ☐Evaluation results with baseline comparisons
- ☐Limitations and failure cases
- ☐One technically challenging feature demonstrated
- ☐Short presentation: problem, approach, results, future work